

Open Materials 2024 (OMat24) Inorganic Materials Dataset and Models

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The discovery and simulation of inorganic materials is core to diverse applications from climate change to semiconductor manufacturing. Artificial intelligence (AI) has the potential to dramatically accelerate materials simulation, discovery and design. Although considerable progress has been made in developing training datasets and machine learning interatomic potential (MLIP) architectures, the state-of-the-art in openly available and reproducible datasets and models lagged behind proprietary models. To address this issue, we present the Open Materials 2024 (OMat24) dataset, comprising over 110M DFT calculations across diverse chemistries, materials, and configurations. MLIP models trained on OMat24 achieve leading performance on the Matbench Discovery leaderboard, surpassing previous models with F1 scores above 0.9 for stability and 20 meV/atom accuracy for formation energy. Models trained on OMat24 also exhibit the highest accuracy in newly developed thermal conductivity and phonon prediction task benchmarks. We show that OMat24’s diversity corrects the consistent softening bias of prior models trained on less diverse datasets, which systematically underpredicted energy, forces, and derivative properties like phonons. The OMat24 dataset has allowed the research community to develop better model architectures, resulting in a step-change improvement inorganic material property prediction accuracy.

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Code: <https://github.com/FAIR-Chem/fairchem>, MIT license

Dataset: <https://huggingface.co/datasets/fairchem/OMAT24>, Creative Commons 4.0 License

Checkpoints: <https://huggingface.co/fairchem/OMAT24>, Custom License

Blogpost: <https://ai.meta.com/blog/fair-news-segment-anything-2-1-meta-spirit-lm-layer-skip-salsa-sona>



1 Introduction

The discovery of new materials lies at the foundation of many pressing global problems. This includes finding new materials for renewable energy storage, producing carbon neutral fuels^{1–3}, or the direct capture of CO₂ from the air⁴, among many others^{5–9}. The search space for new materials is enormous, making the ability to efficiently screen new materials essential. This is challenging for both traditional computational methods and experimental material science methods. Machine Learning (ML) models offer the potential to dramatically improve the efficiency of computational materials simulations compared to traditional methods such as Density Functional Theory (DFT), and address many scientific questions that have been difficult to answer systematically due to their high computational cost.

Recently, there has been a surge in the development of Machine Learning Interatomic Potentials (MLIPs)—surrogates for Density Functional Theory (DFT)—with improving ability to generalize across materials^{10–16}. As in other domains, training large and generalizable models is limited by the availability of large and diverse datasets^{17,18}. The computational materials community has benefited from open *ab-initio* databases such as the Materials Project¹⁹, Alexandria²⁰, OQMD²¹, and AFlow²², which contain near-stable structures across diverse compositions and prototypes. While models trained on these datasets have grown in accuracy and size, they frequently extrapolate poorly to configurations that deviate substantially from the near-equilibrium training data, limiting their usefulness for simulations under relevant conditions—likeley due to insufficiently diverse training data²³. Recent proprietary models and datasets have claimed to address some of these

issues^{10,24}, but without access to the underlying training data, open science efforts cannot benefit from or independently replicate these results.

We introduce the Open Materials 2024 (OMat24) dataset (Figure 1) to improve the generalization of MLIPs across a wide range of materials and to reduce systematic softening. OMat24 contains over 100 million single-point DFT calculations sampled from diverse non-equilibrium atomic configurations and elemental compositions for inorganic bulk materials, building upon public datasets such as MPtrj¹⁶, the Materials Project²⁵, and Alexandria²⁰, which contain equilibrium or near-equilibrium configurations¹⁹. We pretrain several variants of the EquiformerV2^{26,27} and eSEN²⁸ architectures on OMat24 and show that, after fine-tuning to account for differences in DFT settings, state-of-the-art results are achieved on the popular MatBench Discovery leaderboard²⁹. Since its release, all top models^{30–33} on the leaderboard have adopted the dataset. We show that models trained or pretrained on OMat24 display reduced or no systematic softening, and that these improvements are consistent across different model architectures.

2 Results

2.1 The OMat24 dataset

The OMat24 dataset builds upon valuable insights from previous team initiatives such as the Open Catalyst Project¹ and OpenDAC⁴, community contributions like MPtrj¹⁶ and Alexandria²⁰, as well as proprietary releases. Leveraging these learnings, OMat24 was specifically designed to advance MLIP training across inorganic materials and overcome challenges faced by earlier computational materials datasets.

- **Scale:** OMat24 is a large-scale public ab-initio training dataset for inorganic materials up to 2 orders of magnitude larger than other openly available datasets in terms of number of single-point calculations.
- **Diversity:** OMat24 has compositional diversity across the periodic table thanks to community efforts to prototype and catalog hypothetical materials, and contains a large fraction of far-from-equilibrium structures.
- **Open-science:** OMat24 is released under a permissive CC-by license for the benefit of the entire inorganic materials community to build upon.

The OMat24 dataset includes a total of 118 million structures labeled with energy, forces and cell stress. The number of atoms per structure ranges from 1 to 100 atoms per structure. A histogram of the number of atoms per structure is shown in Figure 1b. The majority of OMat24 structures have less than 20 atoms per structure as a direct result of starting from structures in the Alexandria dataset, which are predominantly structures with 16 or less atoms per structure. The OMat24 structures with more than 50 atoms per structure are the larger input structures that were used for AIMD. Details of the sampling strategy and DFT calculations are included in the Methods section.

OMat24 was designed to train MLIP models that predict both equilibrium and non-equilibrium properties. It includes equilibrium and non-equilibrium structures, yielding broader label distributions than relaxation-only datasets. We confirm this by comparing final label distributions to MPtrj and Alexandria, shown in Figure 1b, plotting energies relative to elemental bulk phases computed with each dataset’s DFT settings. OMat24 shows slightly higher energies than Alexandria (the source of input structures) and substantially higher energies than MPtrj, along with notably wider force and maximum cell-stress distributions than both datasets.

OMat24’s elemental coverage spans most of the periodic table (Figure 1c) and closely matches Alexandria and MPtrj (Supplementary Figures B.3, B.4), with oxides over-represented due to their prevalence in open data. Finally, OMat24 exhibits greater diversity of local atomic environments, reflected in higher and more uniform element-pair counts within 3.5 Å (Supplementary Figure B.5), exceeding both MPtrj and Alexandria.

2.1.1 OMat24 train, validation, and test splits

OMat24 is divided into explicit training, validation and test splits to ensure consistent training and evaluation by the community. Training and validation splits are released to allow for model development and iteration. The test set is divided into four different splits. The first split (WBM Test) is to ensure that the training

dataset does not overlap with the Matbench Discovery leaderboard²⁹ created from the WBM dataset³⁴. The other three splits measure the accuracy of the models on in-domain training data (ID) and the ability of the models to generalize to out-of-distribution compositions (OOD-Composition) and elemental compositions (OOD-Element).

The WBM Test split was created using the AFLOW structure prototype labels³⁵ in the AVIARY package³⁶. The prototype label of a structure is a standardized way to classify crystal structures by elemental stoichiometry, space group, Pearson symbol, and Wyckoff positions³⁵. The split includes all OMat24 structures that were generated starting from an Alexandria relaxed structure with a prototype label matching any of prototype labels from the initial or relaxed structures included in the WBM dataset. Additionally, all OMat24 structures with a prototype label matching an initial or relaxed WBM structure are also included in the WBM Test split. The WBM Test split includes a total of 5.3 million structures. Note, filtering the dataset and creating this test split is important to ensure that there was no inadvertent data leakage from the training data to the final Matbench Discovery results, as there is overlap in materials between the Alexandria and WBM datasets.

The OOD-Composition split is constructed by selecting approximately $\sim 5,000$ unique elemental compositions and assigning all structures with matching compositions to the split (573,000 structures). The OOD-Elemental split similarly selects $\sim 3,000$ unique element combinations and includes all matching structures (619,000 structures). A limitation of these splits is the possibility of naive OOD cases (e.g., dilute substitutions or elemental doping) where small stoichiometric changes do not meaningfully increase chemical complexity, though the dataset’s relatively small system sizes and lack of dopants limit such examples. In contrast, the WBM test split—separated by unique anonymized formulas and space groups—consistently yields the highest prediction errors (Supplementary Table H.1), suggesting it represents the most important distribution shift and a more informative test of model generalization.

The training, validation and ID test splits includes all remaining OMat24 structures after creating the 3 test splits described above, and includes a total of 111 million structures. We randomly split this dataset into a training, validation and ID test split containing 100 million, 5 million and 5 million structures respectively for the model training in this work. All dataset split sizes are listed in Table 1.

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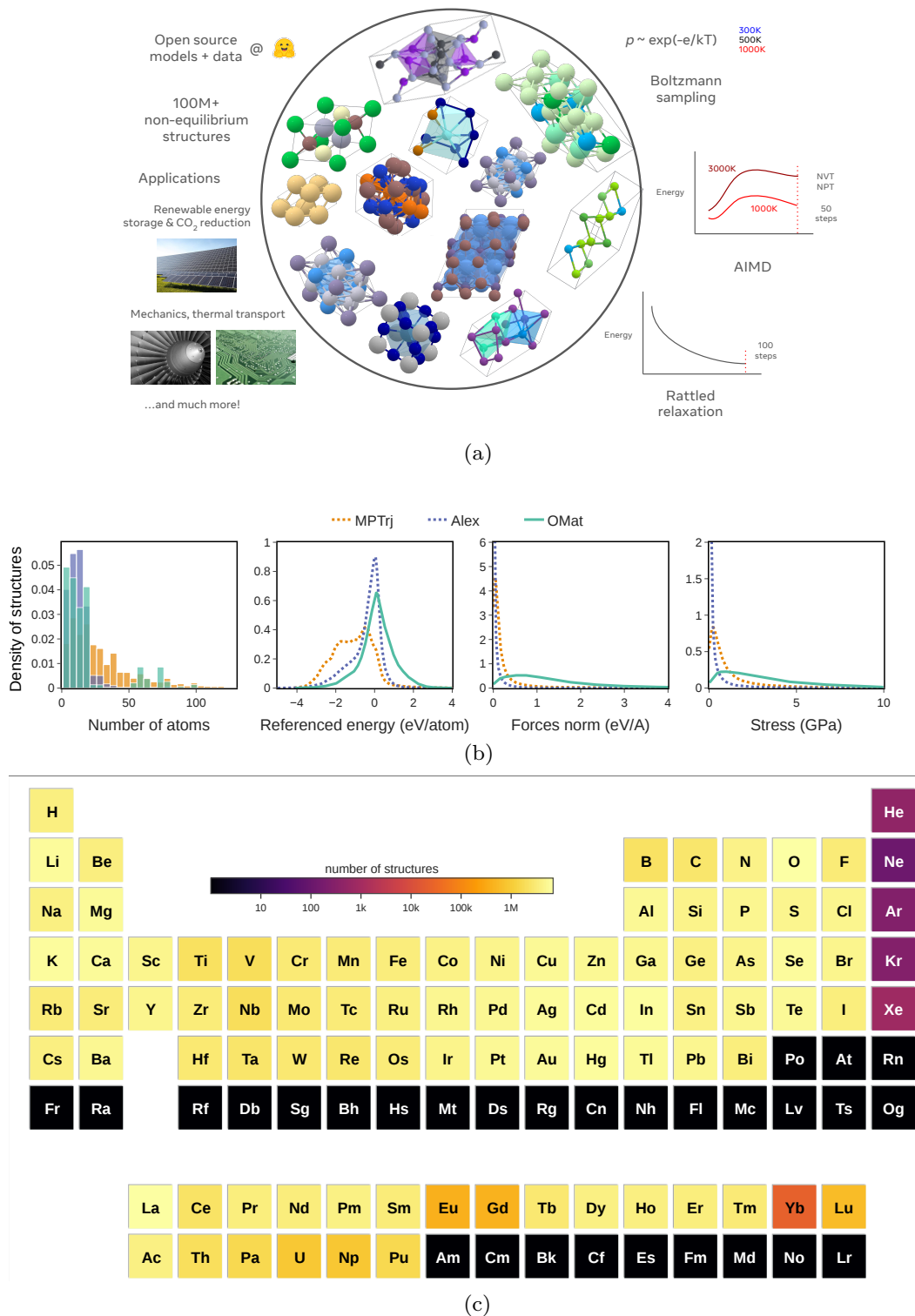


Figure 1 OMat24 dataset overview. (a) Dataset schematic including generation, application areas, and sampling strategies: rattled Boltzmann, rattled relax and AIMD. Structures shown are a random sample across the different sampling strategies. (b) Distribution of frequency density of structures in terms of the number of atoms per structure, the energy per atom referenced with respect to elementary bulk phases, forces norm and max absolute stress element for structures in the MPTrj, Alexandria and OMat24 datasets. (c) Distribution of elements present in the OMat24 dataset

Table 1 OMat24 dataset splits. Size of the OMat24 train, validation and test dataset splits.

Split	Size	Fraction %
Train	100,824,585	85.3
Validation	5,320,549	4.5
WBM Test	5,373,339	4.5
ID Test	5,453,320	4.6
OOD Composition Test	573,301	0.5
OOD Element Test	619,021	0.5

We have also released a development subset of the full dataset, referred to as OMat24-1M, which comprises approximately one million training structures randomly selected from the complete training split. The corresponding validation and test sets contain a number of structures chosen to maintain the same train-to-validation and train-to-test ratios as the full dataset. The OMat24-1M development dataset is particularly useful for rapid training iterations, ablation studies, and hyperparameter optimization.

2.1.2 Dataset limitations

The OMat24 dataset is one of the largest open datasets of its kind for training DFT surrogate models for materials. However, the dataset has limitations similar to many high-throughput datasets that impact the predictions of models trained using the dataset. OMat24 is calculated with PBE and PBE+U levels of DFT, which includes inherent errors in their approximation and resulting calculations³⁷ that are addressed to some extent in other functionals such as PBEsol³⁸, SCAN³⁹, r2SCAN⁴⁰ or hybrid functionals. The OMat24 dataset includes only periodic bulk structures and excludes important effects from point defects, surfaces, non-stoichiometry, and lower dimensional structures. Finally, the OMat24 dataset includes a small fraction of structural relaxations (45,000 total relaxations) starting from distorted relaxed structures in the Alexandria dataset, and does not provide additional or novel information about stable structures.

The calculations in OMat24 differ from those found in the Materials Project PBE and PBE+U calculations. Care must be taken when mixing calculations for analysis or training models. Although the difference in settings is small (pseudopotentials in version 5.4 and the choice of pseudopotential for Yb and W), predictions of total and formation energies can differ. To illustrate these subtleties, we compare calculated formation energies for MP and OMat24 DFT settings in Figure 2 using a set 423 compounds used to fit the MP2020 corrections⁴¹. We use the original MP calculations and compare to calculated energies with OMat24 DFT settings. In order to compute formation energies, we computed elemental references and fit anion and GGA/GGA+U corrections following the methodology used in the MATERIALS2020COMPATIBILITY class in PYMATGEN⁴¹. Additional details about the calculations and mixing corrections are given in Supplementary Section C.

Figure 2 shows parity plots for DFT-predicted formation energies of the 423 compounds. Figure 2b depicts the distribution of differences between MP and OMat24 DFT formation energy predictions. Most differences are under 20 meV/atom and the total mean absolute difference is 13.5 meV/atom, consistent with expected changes from updated pseudopotentials. Deviations above 50 meV/atom mainly arise in GGA+U calculations for magnetic compounds. These differences primarily arise from variations in magnetic ordering: OMat24 initializes all structures as ferromagnetic, whereas some Materials Project (MP) calculations start with lower-energy magnetic states. Consequently, the final magnetic states may differ between the two. The largest differences, marked with red circles in Figure 2b, correspond to compounds that are antiferromagnetic in MP but ferromagnetic in OMat24.

A key limitation in using available DFT datasets to train MLIPs is the aforementioned fixed initialization of magnetic ordering, which does not guarantee that the resulting calculation corresponds to the true magnetic ground state. For instance, models trained on the OMat24 dataset typically predict formation energies for specific magnetic configurations—usually ferromagnetic—rather than the actual magnetic ground states. Accurately computing formation energies of magnetic materials with MLIPs requires careful consideration of the energy differences among various magnetic orderings. Accurate computation of formation energies for magnetic materials using MLIPs requires explicit consideration of the energy differences between possible